

Manifold-Informed Architecture Benchmark

IRIS: 3 classes, 4D input

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waverider @ 4b8002e (-abbrev-re 4b8002ee9a2e3d56a219d7dab695a80b8efd1e07) | 2026-04-14 20:51:52
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Best: Standard (16→8) test acc = 0.9444 ± 0.0157

Manifold: 4D → 3D (25.0% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	IRIS
Input dimensionality	4D
Classes	3
Intrinsic dim (d)	3
Variance threshold (τ)	0.9
Epochs	150
Trials	3

Manifold Discovery

Local PCA over the training set, $k=?$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 3$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	3.2	0.5	2	4	19.0
0.90	2.8	0.4	2	4	29.6
0.85	2.5	0.5	1	3	38.3
0.80	2.1	0.4	1	3	46.2

Per-Class Intrinsic Dimensionality

Class	Mean d	Std	Min	Max
virginica	2.8	0.4	2	3
setosa	2.7	0.5	2	3
versicolor	2.7	0.5	2	3

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Standard (16→8)★	243	0.9444 ± 0.0157	0.1574	3.8866
Wide Manifold (4d→2d→d, d=3)	171	0.9444 ± 0.0157	0.1817	5.5231
Manifold (2d→d, d=3)	63	0.8778 ± 0.0314	0.3069	13.9330
PCA→3D + MLP (2d→d)	57	0.9000 ± 0.0272	0.2405	15.7895
Intrinsic Dim (PCA→3D→output)	24	0.7889 ± 0.0416	0.4703	32.8704

★ = best performing architecture

Key Findings

- **Best architecture:** Standard (16→8) — test accuracy 0.9444 ± 0.0157
- **Manifold compression:** 4D → 3D (25.0% of ambient dimensions carry no structure)

Result Figure

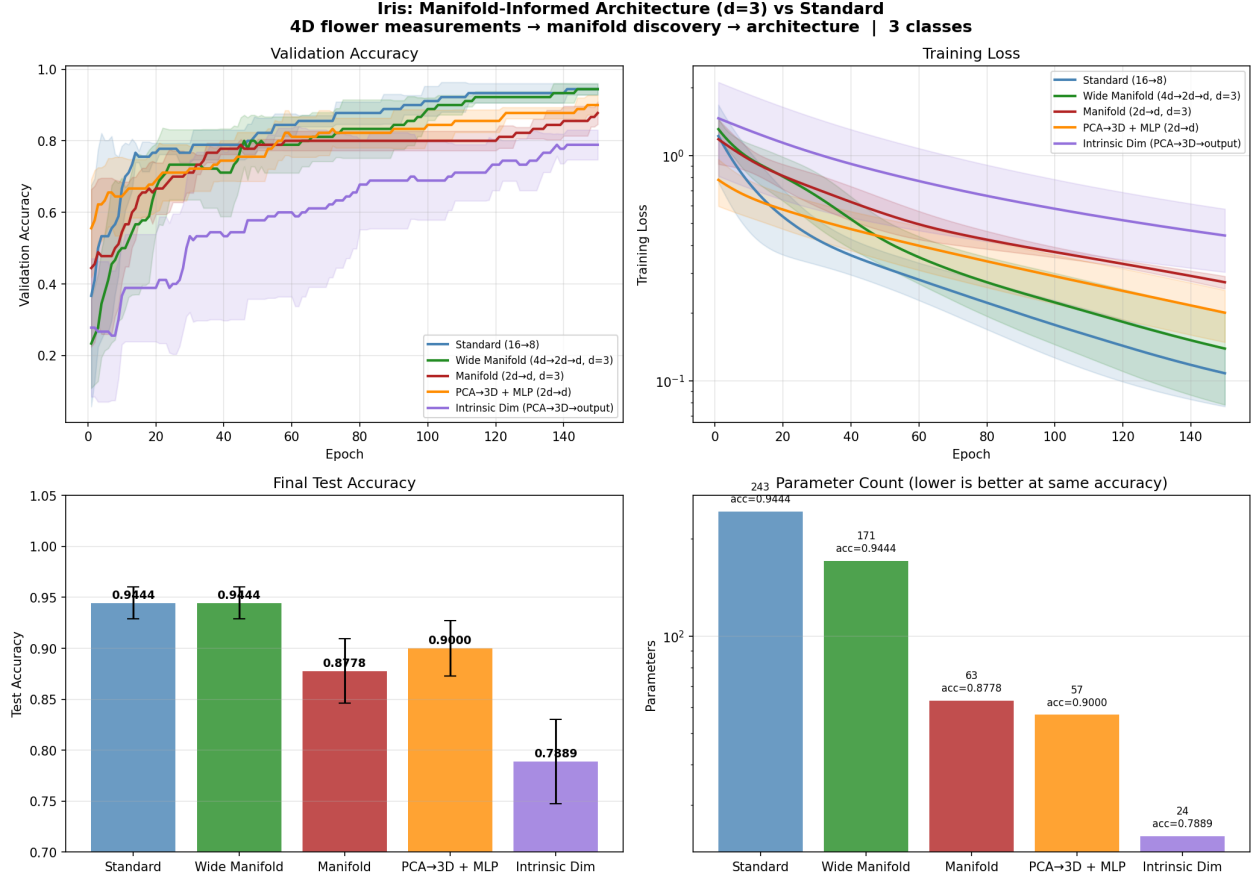


Figure 1: IRIS manifold-informed vs. standard architecture comparison. $d = 3$, 150 epochs, 3 trials.

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	4b8002e (-abbrev-re 4b8002ee9a2e3d56a219d7dab695a80b8efd1e07)
Commit date	2026-04-14 20:51:52 -0400
Commit message	add: cifar10 results
Python	3.12.13
TensorFlow	2.16.2
Device	CPU (forced)
Host	Turing
OS	macOS-26.4-arm64-arm-64bit
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
Generated	2026-04-14 20:54:32
